

Homework 1

EE-556 Mathematics of Data - Fall 2026

In this homework, we consider a multiclass classification task modeled by multinomial (softmax) logistic regression. Your goal will be to analyze the estimator and its properties (convexity, existence/uniqueness), and to derive gradients/Hessians and smoothness bounds. The first part consists of theoretical questions only.

1. Multiclass Softmax Logistic Regression - 30 Points

We now model multiclass classification with classes $c \in \{1, \dots, C\}$. For each sample $(\mathbf{a}_i, b_i)$ with $\mathbf{a}_i \in \mathbb{R}^p$ and $b_i \in \{1, \dots, C\}$, let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_C] \in \mathbb{R}^{p \times C}$ be the class weight matrix. The softmax model defines:

$$\mathbb{P}(b_i = c \mid \mathbf{a}_i) = \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_c)}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)}$$

Assume i.i.d. samples $\{(\mathbf{a}_i, b_i)\}_{i=1}^n$. Our goal is to estimate $\mathbf{X}$ by maximum likelihood (and later with an ℓ_2 regularizer).

(a) (2 points)

Show that the negative log-likelihood f can be written as:

$$\begin{aligned} f(\mathbf{X}) &= -\log \mathbb{P}(b_1, \dots, b_n \mid \mathbf{a}_1, \dots, \mathbf{a}_n) \\ &= \sum_{i=1}^n \left[-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right]. \end{aligned}$$

Solution: We will generally denote the column of the class weight matrix corresponding to the outcome b_i as $\mathbf{x}_{b_i}$. First, we derive the likelihood:

$$L(\mathbf{X}) = \prod_{i=1}^n \mathbb{P}(b_i \mid \mathbf{a}_i) = \prod_{i=1}^n \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_{b_i})}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)}$$

Taking the log we get:

$$\log L(\mathbf{X}) = \log \left\{ \prod_{i=1}^n \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_{b_i})}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)} \right\} = \sum_{i=1}^n \left[\mathbf{a}_i^\top \mathbf{x}_{b_i} - \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right]$$

Negating the expression gives the desired result.

(b) (3 points)

Show that $\mathbf{u} \mapsto \log\left(\sum_{k=1}^C e^{u_k}\right)$ is convex on $\mathbb{R}^C$. Then, show that $f(\mathbf{X})$ is convex. *Hint: use Jensen's inequality.*

Solution: We wish to show that $g : \mathbf{u} \mapsto \log \sum_{k=1}^C e^{u_k}$ is convex on $\mathbb{R}^C$ (where $g : \mathbb{R}^C \rightarrow \mathbb{R}$). For g to be convex on $\mathbb{R}^C$, we need:

$$g(\alpha \mathbf{u}_1 + (1 - \alpha) \mathbf{u}_2) \leq \alpha g(\mathbf{u}_1) + (1 - \alpha) g(\mathbf{u}_2)$$

for all $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^C$ and for all $\alpha \in [0, 1]$.

$$\log \sum_{k=1}^C \exp(\alpha u_{1,k} + (1 - \alpha) u_{2,k}) \leq \alpha \log \sum_{k=1}^C \exp(u_{1,k}) + (1 - \alpha) \log \sum_{k=1}^C \exp(u_{2,k})$$

Exponentiating both sides:

$$\begin{aligned} \sum_{k=1}^C \exp(\alpha u_{1,k} + (1 - \alpha) u_{2,k}) &\leq \exp \left\{ \log \left[\sum_{k=1}^C \exp(u_{1,k}) \right]^\alpha + \log \left[\sum_{k=1}^C \exp(u_{2,k}) \right]^{1-\alpha} \right\} \\ \sum_{k=1}^C \exp(\alpha u_{1,k}) \exp((1 - \alpha) u_{2,k}) &\leq \left[\sum_{k=1}^C \exp(u_{1,k}) \right]^\alpha \left[\sum_{k=1}^C \exp(u_{2,k}) \right]^{1-\alpha} \end{aligned}$$

Dividing both sides:

$$\sum_{k=1}^C \frac{\exp(\alpha u_{1,k})}{\left[\sum_{k=1}^C \exp(u_{1,k}) \right]^\alpha} \cdot \frac{\exp((1 - \alpha) u_{2,k})}{\left[\sum_{k=1}^C \exp(u_{2,k}) \right]^{1-\alpha}} \leq 1$$

Recognising the Softmax pmf, this becomes:

$$\begin{aligned} \sum_{k=1}^C \mathbb{P}(b_i = k \mid \mathbf{u}_1)^\alpha \mathbb{P}(b_i = k \mid \mathbf{u}_2)^{1-\alpha} &\leq 1 \\ \sum_{k=1}^C \mathbb{P}(b_i = k \mid \mathbf{u}_1) \frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)^{1-\alpha}}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)^{1-\alpha}} &\leq 1 \end{aligned}$$

Rewriting as an expectation:

$$E_{b_i | \mathbf{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)} \right]^{1-\alpha} \right\} \leq 1$$

By Jensen's inequality:

$$E_{b_i | \mathbf{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)} \right]^{1-\alpha} \right\} \leq \left\{ E_{b_i | \mathbf{u}_1} \left[\frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)} \right] \right\}^{1-\alpha}$$

This is because $y = x^\theta$ is concave by the definition of a convex function using the positive semi-definiteness of the Hessian (which here reduces to a scalar):

$$\frac{\partial}{\partial x} y = \theta x^{\theta-1}, \quad \frac{\partial^2}{\partial x^2} y = \theta(\theta - 1) x^{\theta-2} < 0$$

for any $\theta \in [0, 1]$ (with $\theta = 1 - \alpha$). The RHS of the Jensen inequality is:

$$\begin{aligned} \left\{ E_{b_i|\mathbf{u}_1} \left[\frac{\mathbb{P}(b_i = k | \mathbf{u}_2)}{\mathbb{P}(b_i = k | \mathbf{u}_1)} \right] \right\}^{1-\alpha} &= \left(\sum_{k=1}^C \frac{\mathbb{P}(b_i = k | \mathbf{u}_2)}{\mathbb{P}(b_i = k | \mathbf{u}_1)} \mathbb{P}(b_i = k | \mathbf{u}_1) \right)^{1-\alpha} \\ &= \left(\sum_{k=1}^C \mathbb{P}(b_i = k | \mathbf{u}_2) \right)^{1-\alpha} = 1 \end{aligned}$$

since it is a pmf. This implies that:

$$E_{b_i|\mathbf{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k | \mathbf{u}_2)}{\mathbb{P}(b_i = k | \mathbf{u}_1)} \right]^{1-\alpha} \right\} \leq 1$$

(c) (1 point)

Since the negative log-likelihood f is convex, every local minimizer is a global minimizer. The maximum likelihood estimation problem is

$$\mathbf{X}_{ML}^* = \arg \min_{\mathbf{X} \in \mathbb{R}^{p \times C}} f(\mathbf{X})$$

But does f always attain its infimum? The following three questions examine this issue.

Explain the difference between infima and minima. Give an example of a convex function on $\mathbb{R}$ that does not attain its infimum.

Let $f: \mathcal{X} \rightarrow \mathbb{R}$ be a function defined on a domain $\mathcal{X}$. The infimum of f , denoted $\inf_{x \in \mathcal{X}} f(x)$, is the greatest lower bound of its image $f(\mathcal{X})$. Formally, a real number $m = \inf_{x \in \mathcal{X}} f(x)$ if $m \leq f(x)$ for all $x \in \mathcal{X}$, and for every $\epsilon > 0$, there exists $x \in \mathcal{X}$ such that $f(x) < m + \epsilon$.

A minimum is an infimum that is explicitly attained by the function within its domain.

Specifically, the infimum is a minimum, denoted $\min_{x \in \mathcal{X}} f(x)$, if and only if $\exists x^* \in \mathcal{X}$ such that $f(x^*) = \inf_{x \in \mathcal{X}} f(x)$. Consequently, all minima are infima, but an infimum is a minimum strictly when the infimum is an element of the image set $f(\mathcal{X})$.

To illustrate a convex function on $\mathbb{R}$ that does not attain its infimum, consider $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = e^x$. This function is strictly convex and bounded below by 0. Evaluating the asymptotic behavior yields $\lim_{x \rightarrow -\infty} e^x = 0$, establishing that $\inf_{x \in \mathbb{R}} e^x = 0$. However, because $\nexists x \in \mathbb{R}$ such that $e^x = 0$, the infimum is not attained in the domain, and therefore $\min_{x \in \mathbb{R}} e^x$ is undefined.

(d) (2 points)

Assume there exists $\mathbf{X}_0 \in \mathbb{R}^{p \times C}$ such that for all i ,

$$\mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} > 0.$$

This is called one-versus-all complete separation in multiclass settings. Give a geometric interpretation (e.g., for $p = 2$) and explain why the name is appropriate.

GEOMETRIC INTERPRETATION IS MISSING

The name "one-versus-all complete separation" is appropriate because the inequality mathematically compares the score of the correct class ("one") against the maximum score of all alternative classes ("all").

(e) (3 points)

Complete separation allows perfect classification of the training data. However, as you will show next, it prevents the existence of a finite maximum likelihood estimate.

In a one-versus-all complete separation setting (as in (d)), prove that f does not attain its minimum. Hint: consider $f(\alpha \mathbf{X}_0)$ as $\alpha \rightarrow +\infty$ and compare it to $f(\mathbf{X}_0)$.

By assumption, $\mathbf{X}_0$ achieves complete separation, that is, for all i ,

$$\mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} > 0.$$

It is straightforward to check that if $\mathbf{X}_0$ achieves complete separation, then $\alpha \mathbf{X}_0$ achieves complete separation as well for any $\alpha > 0$:

$$\mathbf{a}_i^\top (\alpha \mathbf{x}_{0,b_i}) - \max_{k \neq b_i} \mathbf{a}_i^\top (\alpha \mathbf{x}_{0,k}) > 0 \iff \alpha \left(\mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} \right) > 0 \iff \mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} > 0$$

To evaluate the behavior of the negative log-likelihood under complete separation, we substitute the scaled weight matrix $\alpha \mathbf{X}_0$ into the objective function:

$$f(\alpha \mathbf{X}_0) = \sum_{i=1}^n \left[-\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i} + \log \sum_{k=1}^C \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}) \right]$$

We can factor out the exponential term corresponding to the true class b_i from the sum inside the logarithm:

$$\begin{aligned} \sum_{k=1}^C \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}) &= \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i}) \left(1 + \sum_{k \neq b_i} \exp(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i})) \right) \\ \log \sum_{k=1}^C \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}) &= \alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i} + \log \left(1 + \sum_{k \neq b_i} \exp(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i})) \right) \end{aligned}$$

Substituting this back into $f(\alpha \mathbf{X}_0)$, the $-\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i}$ and $+\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i}$ terms cancel out, yielding:

$$f(\alpha \mathbf{X}_0) = \sum_{i=1}^n \log \left(1 + \sum_{k \neq b_i} \exp(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i})) \right)$$

The complete separation assumption guarantees that $\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i} > \alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}$ for all $k \neq b_i$. Consequently, the exponent is strictly negative for all incorrect classes:

$$\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i}) < 0$$

As $\alpha \rightarrow +\infty$, the exponential of a strictly negative number approaches zero:

$$\lim_{\alpha \rightarrow +\infty} \exp(\alpha(\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i})) = 0$$

Evaluating the limit for the entire function gives:

$$\lim_{\alpha \rightarrow +\infty} f(\alpha \mathbf{X}_0) = \sum_{i=1}^n \log(1 + 0) = 0$$

This shows that the infimum of $f(\mathbf{X})$ is 0.

However, for any finite weight matrix $\mathbf{X} \in \mathbb{R}^{p \times C}$, the sum $\sum_{k \neq b_i} \exp(\mathbf{a}_i^\top \mathbf{x}_k - \mathbf{a}_i^\top \mathbf{x}_{b_i})$ is strictly greater than 0 because the exponential function never evaluates to zero. Therefore, $f(\mathbf{X}) > 0 \forall \mathbf{X} \in \mathbb{R}^{p \times C}$. Because the function can approach 0 asymptotically but never reach it for any finite parameters, it does not attain its minimum.